



Aggregates – Properties, Types & Uses

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Aggregate - Definition

- Aggregate are the important constituent in concrete. Aggregate are granular material, derived from the most part from the natural rocks, crushed stones, or natural gravels and sands.
- Aggregate generally occupy about 70% to 80% of the volume of concrete and can therefore be expected to have an important influence on its properties.





Aggregate - Requirements

An ideal aggregate used for the manufacturing of concrete and mortar, should meet the following requirements.

1. It should consist of natural stones, gravels and sand or in various combinations of these materials.
2. It should be hard, strong and durable.
3. It should be dense, clear and free from any coating.
4. It should be free from injurious vegetable matters.
5. It should not contain flaky (angular) and elongated pieces.
6. It should not contain any material liable to attack steel reinforcement in case of reinforced concrete.



Aggregate - Classification

CLASSIFICATION BASED ON SIZE:

☐ Natural aggregates:

This kind of aggregate is taken from natural deposits without changing their nature during the process production such as crushing and grinding.

☐ Manufactured (synthetics) aggregates:

This is a kind of man-made materials produced as a main product or an industrial by-product. Some example are blast furnace slag, air cooled slag and broken bricks. Synthetics aggregates are produced by thermally processed materials such as expanded clay and shale.



Aggregate - Classification

CLASSIFICATION BASED ON SIZE:

❑ Coarse aggregate:

Aggregate which retained on the No.4 (4.75mm) sieve. The function of the coarse aggregate is to act as the main loadbearing component of the concrete.



❑ Fine aggregate:

Aggregate passing No.4 (4.75mm) sieve and predominately retained on the No.200 (75 μ) sieve. The fine aggregate serve the purpose of filling all the open space in between the coarse particles.





Aggregate - Classification

CLASSIFICATION BASED ON SHAPE:

- The shape of aggregates is an important characteristic, since it affects the workability of concrete.

CLASSIFICATION	EXAMPLE
Rounded	River or seashore gravels
Partly rounded	Pit sands & Gravels
Angular	Crushed Rocks
Flaky	Laminated rocks



FLAT



ELONGATED



ANGULAR



ROUND



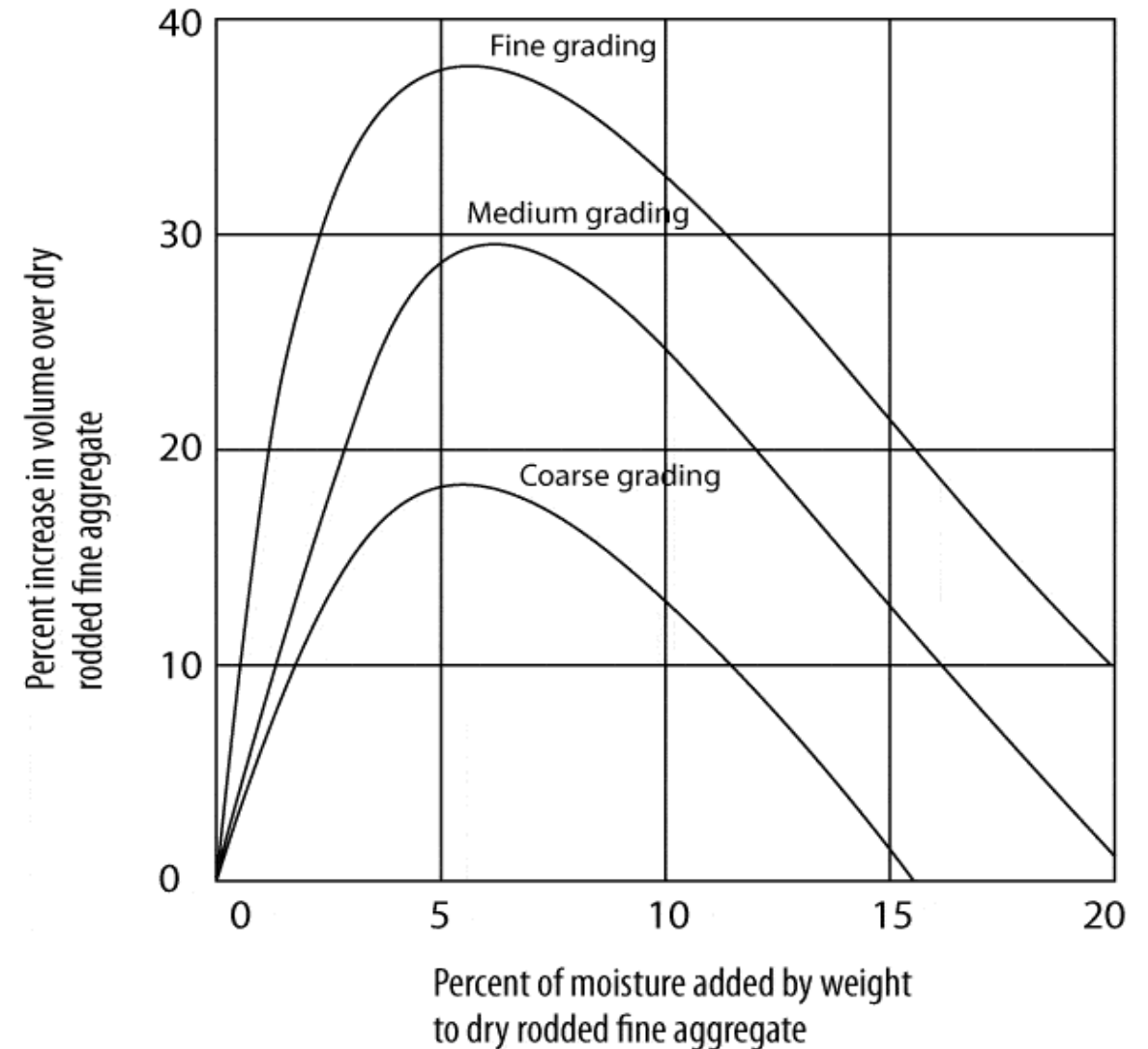
Bulking of Sand

- ❖ Bulking of sand refers to the increase in volume of sand due to the presence of moisture.
- ❖ When dry sand comes into contact with water, a thin film of water forms around the sand particles, causing them to separate.
- ❖ This results in an expansion of the sand volume, known as bulking.
- ❖ The bulking of sand phenomenon is significant in construction, especially in activities like concrete mix proportioning.
- ❖ Engineers and builders need to account for the bulking factor when determining the correct amount of sand to be used in construction materials.



Bulking of Sand

Moisture Content in Sand	Increase in Volume of Sand
2%	15%
3%	20%
4%	25%
5%	30%





Properties of Aggregate

Physical Properties

- **Specific Gravity:** 2.6–2.9.
- **Bulk Density:** 1400–1600 kg/m³ (loose); 1600–1800 kg/m³ (compacted).
- **Water Absorption:** < 2% (good quality).
- **Surface Texture:** Rough (crushed stone) → better bond; smooth (river gravel) → less bond.
- **Grading:** Well-graded aggregates improve density and reduce voids.

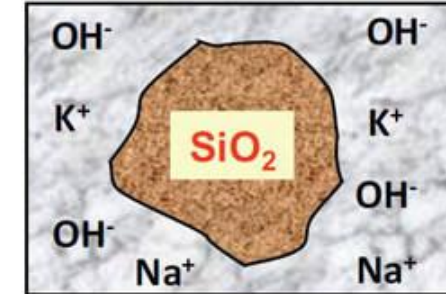
Mechanical Properties

- **Crushing Strength:** Aggregate Crushing Value (ACV) should be < 30% for concrete.
- **Toughness:** Aggregate Impact Value (AIV) should be < 30%.
- **Hardness:** Los Angeles Abrasion Value < 30% for wearing surfaces.
- **Elastic Modulus:** Stiffer aggregates increase concrete stiffness.

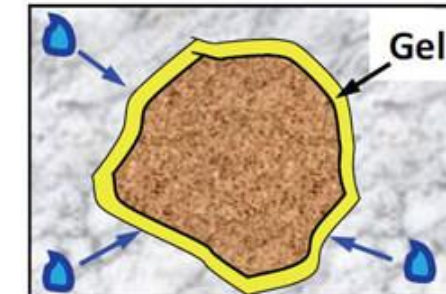


Alkali Aggregate Reaction

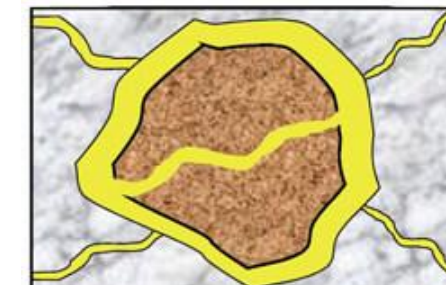
Alkali-Aggregate Reaction (AAR) is a deleterious chemical process in concrete where alkaline hydroxides from cement paste react with reactive minerals in the aggregates. Over time, this reaction produces a substance that absorbs moisture, expands, and exerts internal pressure, leading to severe cracking, spalling, and loss of structural strength. Because of its slow, destructive nature, it is often referred to as "concrete cancer"



Reaction between the alkali hydroxides (Na, K, & OH) from the cement and unstable silica, SiO_2 in some types of aggregate.



The reaction produces an alkali-silica gel, which absorbs water from the surrounding paste...



...and expands.

The internal expansion eventually leads to cracking of the surrounding concrete



Examples of Alkali Aggregate Reaction





Manufactured Sand

- ❖ **Manufactured Sand (M-Sand)** is a fine aggregate produced by **crushing, shaping, and screening hard rocks or stones** in a controlled manufacturing process to obtain particles of sand-sized dimensions.
- ❖ Unlike natural sand, which is obtained from **riverbeds, pits, or other natural deposits**, M-sand is deliberately manufactured to achieve suitable particle size and grading.





Ideal Properties of Manufactured Sand

- **Well-graded particles** with a proper distribution of different particle sizes to improve packing and reduce voids.
- **Cubical and reasonably angular particles** that provide good interlocking and bonding with cement paste.
- **Free from harmful impurities** such as clay lumps, organic matter, coal, lignite, excessive mica, salts, and other harmful substances.
- **Controlled fines content** to avoid excessive water demand and loss of workability.
- **Hard, strong, and durable particles** obtained from suitable parent rock resistant to weathering and abrasion.
- **Good durability** without materials that may cause chemical deterioration, alkali–aggregate reactions, excessive shrinkage, or other durability problems.
- **Consistent quality** with controlled grading, particle shape, fines content, and overall properties during manufacturing.



Natural vs Manufactured Sand

Property	Manufactured Sand (M-Sand)	Natural Sand
Source	Produced by crushing suitable rocks	Obtained from natural deposits such as rivers and pits
Production	Artificially manufactured	Naturally formed by weathering and erosion
Particle shape	Generally angular/cubical	Generally rounded to sub-rounded
Surface texture	Usually rough	Usually comparatively smooth
Grading	Can be controlled during manufacturing	Depends on natural deposit and processing
Fines	Generally higher and need to be controlled	Usually lower, depending on source
Workability	May require more water/admixture because of angular and rough particles	Generally, gives better workability due to smoother, rounded particles



Natural vs Manufactured Sand

Property	Manufactured Sand (M-Sand)	Natural Sand
Bond with cement paste	Generally good because of rough/angular surface	Comparatively lower mechanical interlocking because of rounded particles
Quality consistency	Can be relatively consistent with controlled production	Can vary significantly from one natural source to another
Environmental impact	Reduces dependence on river sand; uses crushed rock	Excessive river sand extraction can damage river ecosystems
Availability	Can be produced near suitable rock sources	Depends on availability of suitable natural deposits
Particle shape control	Can be controlled using suitable crushing/shaping equipment	Cannot be directly controlled during natural formation
Cost	Depends on quarry, processing and transportation	Depends strongly on availability and transportation
Construction use	Concrete, mortar, plastering and other applications when conforming to specifications	Concrete, mortar, plastering and other applications when conforming to specifications

THANK
YOU